
Sensor-Stream Safety Bands: An Assurance Case

Top claim: the humanoid stays within safe bands during contact

Method: claim-argument-evidence

Scope: one featured sensor stream

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Abstract.

This template states an assurance case for a humanoid’s sensor-stream safety bands: a top claim, the arguments that support it, and the evidence each argument rests on.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords

1 Top claim and scope

The top claim is stated and bounded before any argument is offered.

Top claim: the humanoid’s live safety margins remain within the governing gate bounds
Scope: the featured sensor stream under nominal and fault conditions
Assumption: sensors are calibrated and the seed is fixed

2 Hazards

The case addresses three hazards.

- H1 contact force exceeds the budget
- H2 clearance falls below the minimum
- H3 the system fails to reach a safe state on fault

3 Argument

Each hazard is discharged by a claim-argument-evidence chain.

1. **Claim.** margins stay within bounds for H1-H3
2. **Argument.** each gate bound is monitored live and forces a safe state on breach
3. **Evidence.** sweep results and the sensor-band record support the bounds

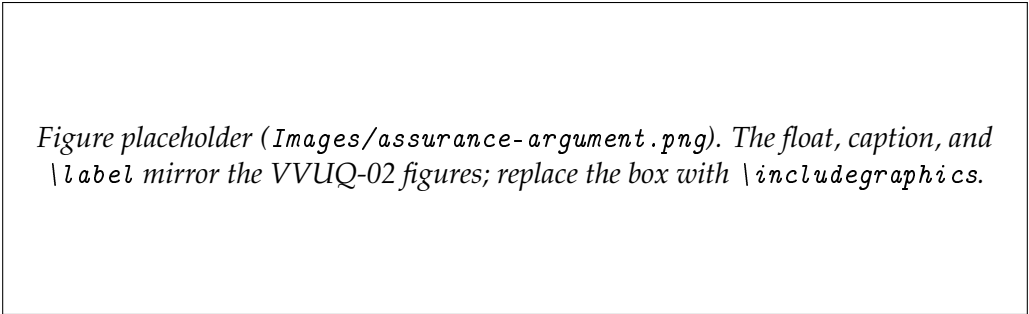


Figure 1: Figure 1. Placeholder assurance-argument structure (claim to evidence).

4 Evidence

Claim	Argument	Evidence
H1 force	live force gate	band within budget
H2 clearance	live clearance gate	band above minimum
H3 fault	e-stop to safe state	fault test passes

Table 1: Table 1. Claim-argument-evidence summary (modeled on VVUQ-05 Table 1).

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Ethical Disclosures

The author declares no competing interests. This is an independent draft and is not endorsed, sponsored, or approved by any trial sponsor, CRO, site, IRB, regulator, or medical society, nor by CFR, ICH, or FDA. All supporting numbers are simulation or illustrative results.

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Data Availability

Supporting records are openly available in the kevinkawchak/cancer-automated and kevinkawchak/physical-ai-oncology-trials GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

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